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In [6]: import pandas as pd
import matplotlib.pyplot as plt
import cartopy.crs as ccrs

# 1. 读入数据
eq = pd.read_csv("C:\\Users\\Libing\\Desktop\\usgs_earthquakes.csv",
                 parse_dates=['time', 'updated'],
                 index_col='id')

# 2. 选出震级最大的 50 次地震
top50 = eq.nlargest(50, 'mag')

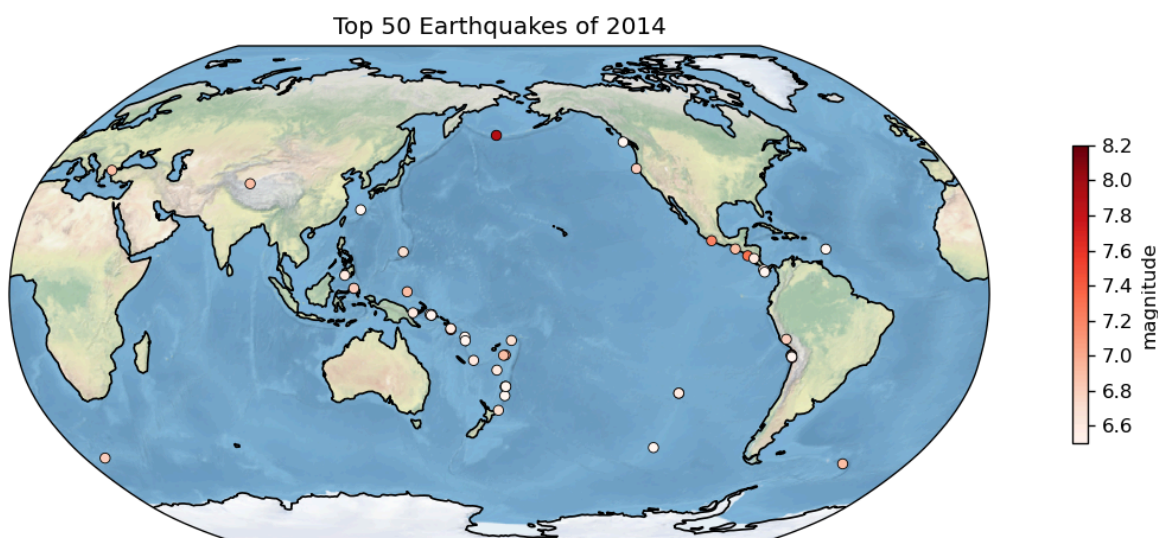
# 3. 画图: Robinson 投影, 中心经度 180°
plt.figure(figsize=(12, 4), dpi=120)
ax = plt.axes(projection=ccrs.Robinson(central_longitude=180))
ax.stock_img()          # 背景地球图
ax.coastlines()         # 海岸线

# 散点: 颜色代表震级
sc = ax.scatter(top50['longitude'],
               top50['latitude'],
               c=top50['mag'],
               s=26, #点的大小
               zorder=2,
               transform=ccrs.PlateCarree(),
               cmap='Reds',
               edgecolors='k',
               linewidths=0.5)

# 颜色条
cbar = plt.colorbar(sc, ax=ax, orientation='vertical', label='magnitude', shrink=0.
                    ticks=[6.6,6.8,7.0,7.2,7.4,7.6,7.8,8.0,8.2])

ax.set_title('Top 50 Earthquakes of 2014')
plt.tight_layout()
plt.show()

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In [35]: import xarray as xr
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import cartopy.feature as cfeature

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# 中文字体设置, 防止中文乱码和负号变方块
plt.rcParams['font.sans-serif'] = ['SimHei', 'Microsoft YaHei', 'SimSun']
plt.rcParams['axes.unicode_minus'] = False

# 读取数据
ds = xr.open_dataset(r"C:\Users\Libing\Desktop\NCALDAS_NOAH0125_Trends.A198010_
da = ds["Trend_Tair_f"].squeeze() #提取所需数据
lon = ds["lon"]
lat = ds["lat"]

# 样点位置
lon0, lat0 = -100, 35

# 2.1 全局图
fig, ax = plt.subplots(figsize=(9, 4), subplot_kw={"projection": ccrs.Robinson()})
ax.set_global() #显示全局地图

# 绘制地理特征: 陆地、海洋、海岸线
ax.add_feature(cfeature.LAND, edgecolor="k", facecolor="lightgray")
ax.add_feature(cfeature.OCEAN, facecolor="lightblue")
ax.coastlines()

# 添加网格线和经纬度标签
gl = ax.gridlines(draw_labels=True, linestyle="--", linewidth=0.3)
gl.top_labels = False #取消顶部的经度标签
gl.right_labels = False #取消右侧的维度标签

# 填色图 and 颜色条
im = ax.pcolormesh(lon, lat, da, transform=ccrs.PlateCarree(), cmap="coolwarm")
plt.colorbar(im, ax=ax, pad=0.02, label="气温趋势 Trend_Tair_f (K / 十年)")

# 标注样点
ax.plot(lon0, lat0, "ko", transform=ccrs.PlateCarree(), label="样点") # 作为 Legend
ax.text(lon0, lat0, " 样点", transform=ccrs.PlateCarree(), ha="left", va="center")

# 图例
ax.legend(loc="lower left", fontsize=8)

# 添加坐标轴标签 and 标题
ax.set_xlabel("经度")
ax.set_ylabel("纬度")
ax.set_title("全球气温趋势 Trend_Tair_f")

#显示图形
plt.tight_layout()
plt.show()

# 2.2 区域图 (这里以美国本土作为例子)
fig, ax = plt.subplots(figsize=(7, 4),
    subplot_kw={"projection": ccrs.LambertConformal( # 另一种投影方法: Lambert
        central_longitude=-97, central_latitude=35)})

# 区域范围: CONUS (美国本土)
ax.set_extent([-125, -67, 25, 53], crs=ccrs.PlateCarree())

# 绘制地理特征: 陆地、海洋、州界、海岸线
ax.add_feature(cfeature.LAND, edgecolor="k", facecolor="lightgray")
ax.add_feature(cfeature.OCEAN, facecolor="lightblue")
ax.add_feature(cfeature.STATES, linewidth=0.4)

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ax.coastlines()

# 添加网格线和经纬度标签
gl = ax.gridlines(draw_labels=True, linestyle="--", linewidth=0.3)
gl.top_labels = False
gl.right_labels = False

# 区域填色图和颜色条
im = ax.pcolormesh(lon, lat, da, transform=ccrs.PlateCarree(), cmap="coolwarm")
plt.colorbar(im, ax=ax, pad=0.02, label="气温趋势 Trend_Tair_f (K / 十年)")

# 样点和注释
ax.plot(lon0, lat0, "ko", transform=ccrs.PlateCarree(), label="样点")
ax.text(lon0, lat0, " 样点", transform=ccrs.PlateCarree(), ha="left", va="center")
ax.legend(loc="lower left", fontsize=8)

# 添加文本框
ax.text(
    0.02, 0.98, "区域: 美国本土 (CONUS) \n变量: Trend_Tair_f",
    transform=ax.transAxes, ha="left", va="top", fontsize=8,
    bbox=dict(boxstyle="round", facecolor="white", alpha=0.7))

# 坐标轴标签和标题
ax.set_xlabel("经度")
ax.set_ylabel("纬度")
ax.set_title("美国本土气温趋势 Trend_Tair_f")

plt.tight_layout()
plt.show()

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